PLAN-02 - Problem and Current State

POS-DGFY - Global Location Switching | Plan v1.0 | Revision 2026-09-27

Field	Value
Project	POS-DGFY - Global Location Switching
Plan version	1.0
Section	PLAN-02 - Problem and Current State
Revision date	2026-09-27
Plan status	NOT EXECUTION READY
Purpose	Record evidence about the current architecture, the actual gap, and constraints that must not be guessed.
Dependencies	PLAN-01; AGENTS.md; docs/START_HERE.md; architecture boundary and governance docs; current source tree.

Problem statement

The POS needs one understandable location context, but operating a branch and merely viewing branch data have different safety requirements. A free global switch would risk cross-branch sales, stock, cash, queue, and audit errors; a Shift-only entry point is difficult to discover.

Confirmed current state

Area	Evidence	Classification
Permission	apps/dgfy-api/src/config/permissions.js:115 defines pos:switch_location.	REUSE
Operating transition	apps/dgfy-api/src/routes/pos.js:343 exposes the governed shift switch endpoint.	REUSE
Frontend service	packages/web-core/src/features/pos/services/posService.js:686 posts the transition.	REUSE
Context state	TerminalPage.jsx:720-722 tracks operating, viewing, and queue scopes.	EXTEND
Admin viewing	TerminalOperationsWorkspace.jsx:768-839 exposes Register Branch and Viewing Branch controls.	EXTEND
Shift UI	TerminalOperationsWorkspace.jsx:1131+ contains Change Active Shift Location.	REUSE
Inventory authority	ADR 0009 identifies item_location_stocks and location movements as stock truth.	LEAVE UNCHANGED

Current checkout baseline

- Branch: feat/DGFY-343-2067-pos-workflow-consolidation.
- Baseline commit observed during planning: 786cd4e8329db1a65398ce6aa7eb020ffc8f3b28.
- The working tree is dirty with unrelated application, generated, and local skill changes. Implementation must preserve them and must use a clean, explicitly selected worktree or branch.

Root cause and secondary effects

The root cause is not the absence of location logic. The system already has partial location context and a governed shift transition. The gap is that the context is not yet presented and reconciled as one cross-module product contract, and not every requested module has a confirmed scope contract.

- Secondary effect: users may interpret viewing a branch as operating that branch.
- Secondary effect: adding a global selector without request invalidation could display stale data from the previous branch.
- Secondary effect: modules with unclear scope could either leak data or unnecessarily block valid operations.

Documentation constraint

The main implementation phase ledger currently has a high-water mark below the location-access documents, while those documents define Phase 354 and runtime Phases 355-357. Reconcile the ledger before implementation; do not invent a new phase sequence.

Stakeholders

Stakeholder	Need	Risk
Cashier	Fast, obvious branch context and safe switch flow.	Accidental sale or cash assignment to wrong branch.
Branch manager	Location-scoped operations, staff, stock, reports, and settings.	Unauthorized cross-location mutation.
Owner / administrator	Authorized cross-location viewing and controlled administration.	Overbroad access and confusing operating state.
Finance / audit	Stable historical reports and traceable transitions.	Rewritten history or missing actor attribution.
Engineering / operations	Observable, reversible rollout and clear recovery.	Partial transitions and difficult incident repair.